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Quantification of Hyperreflective Foci in Age-related Macular Degeneration by Polarization-Sensitive OCT

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Purpose: To quantify hyperreflective foci (HRF) using polarization-sensitive OCT (PS-OCT) and evaluate the effect of anti-VEGF therapy on HRF reduction.

Design: Retrospective study.

Subjects: Twelve eyes from 12 patients with neovascular age-related macular degeneration (nAMD) who underwent PS-OCT imaging before and after 4 intravitreal injections of faricimab, administered every 4 weeks.

Methods: Retinal layers between the inner border of the inner nuclear layer and the outer border of the outer segment of photoreceptors were analyzed. Regions with entropy values exceeding the mean + 2 standard deviations from healthy controls (entropy: 0.0896 ± 0.0270) were isolated to quantify hyperentropic foci (HEF), representing HRF.

Main Outcome Measures: The sum and area of HEF were calculated as total HEF and area aggregate to compare before and after the treatment.

Results: The mean patient age was 77.4 ± 8.6 years, with an equal distribution of males and females (6 each). Total HEF was significantly associated with worse baseline visual acuity (coefficient: 0.0015297, $P = 0.0004$). Faricimab treatment significantly reduced both total HEF ($P = 0.0049$) and area aggregate ($P = 0.0068$). However, conventional OCT did not detect significant improvements in total reflectivity nor area aggregate.

Conclusions: Polarization-sensitive OCT is a valuable imaging modality for extracting and quantifying HRF in the neurosensory retina. Faricimab significantly reduces HEF in nAMD eyes, highlighting its potential therapeutic effect beyond conventional OCT-detected changes.

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Age-related macular degeneration (AMD) is a leading cause of blindness among the elderly population, particularly in industrialized countries.[1] The development and progression of AMD have been well characterized in the literature, detailing the clinical features of early, intermediate, and late stages.[2] In the late stage of AMD, macular neovascularization drives the pathology of neovascular AMD (nAMD), causing severe vision loss by subretinal hemorrhage and fluid leakage.[3]

Hyperreflective foci (HRF) are clinical signs observed on OCT images. Hyperreflective foci are defined as discrete well-demarcated dots in neurosensory retina, with reflectivity greater than or equivalent to retinal pigment epithelium (RPE). Hyperreflective foci are observed approximately 40% of eyes with nAMD and are thought to represent the migration of RPE cells or inflammatory cells into neurosensory retina.[4–7] Some of HRF lesions are consider as precursors of hard exudates, consisting of extravasated lipids,[8,9] because they are also observed in retinal vascular diseases including diabetic macular edema and retinal vein occlusion.[10,11] Recent studies have identified HRF as well as drusen as biomarkers of AMD progression, with their presence correlating with disease severity.[12–14] Thus, there is an increasing need to visualize and quantify HRF objectively to assess disease activity and predict prognosis on an individual basis.

Although conventional OCT provides detailed characteristics of HRF and their relationship with AMD prognosis, it is challenging to quantify HRF objectively. This is partly because the reflectivity of some HRF is indistinguishable from that of existing retinal layers, such as retinal ganglion cell layer and RPE. Therefore, an imaging modality with higher sensitivity is needed for accurate quantitative analysis.

Polarization-sensitive OCT (PS-OCT) is a promising imaging technique, which can offer new insights into the

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pathology of retinal diseases. Polarization-sensitive measures the polarization properties of light within tissues, providing tissue-specific contrast for each retinal layer.[15] Similar to conventional OCT, PS-OCT produces cross sectional and 3-dimensional images, but it also reveals material distribution based on polarimetric entropy. In the retina, PS-OCT detects high polarimetric entropy in the RPE because of intracellular melanin, distinguishing it from other layers under normal physiological conditions.[16,17] Moreover, under pathological conditions, PS-OCT can detect collagen fibers, lipids, hard exudates, and activated inflammatory cells—that is, elements considered to contribute to HRF composition.[18–20]

Given these capabilities, we hypothesized that PS-OCT is a suitable modality for detecting HRF. In this study, we quantified HRF by measuring entropy values with PS-OCT and examined whether anti-VEGF therapy improves both intraretinal HRF and exudative changes in nAMD. Our findings aim to validate the ability of PS-OCT to detect HRF accurately. This study PS-OCT will contribute to a deeper understanding of AMD disease pathology and introduce a novel biomarker for characterizing HRF in nAMD.

Methods

Study Design

This study used data collected from a retrospective study. The protocol was approved by the institutional review board of the University of Tokyo (protocol #11822) and adhered to the principles of the Declaration of Helsinki. Written informed consent was obtained from all participants.

Participants

Patients aged ≥50 years, diagnosed with any types of nAMD and treated with 4 initial every-4-week intravitreal injections of faricimab were included in this study. Best-corrected visual acuity (BCVA; logarithm of the minimum angle of resolution converted from Snellen), color fundus photography, swept-source (SS)-OCT, OCT angiography, fluorescein angiography, and indocyanine green angiography, and were used to diagnose nAMD and classify nAMD subtypes. Neovascular age-related macular degeneration was categorized into macular neovascularization type 1 (occult), type 2 (classic), type 3 (retinal angiomatous proliferation), and polypoidal choroidal neovasculopathy. Central subfield thickness was defined as the average thickness of the circular 1-mm diameter circular area of the fovea. The patients with a history of diabetic retinopathy, glaucoma treatments, other retinochoroidal diseases, or prior intraocular surgery (except for cataract surgery) were excluded from the analysis.

PS-OCT and Imaging Analysis

We used commercially available PS-OCT (ROCTIA, Tomey Corporation), as previously used in our study.[21] The mechanical principle of PS-OCT system is described in Figure S1 (available at www.ophthalmologyscience.org) besides our previous reports.[22,23] For each scan, a 6 mm × 6 mm macular area was covered with a raster scan pattern consisting of 512 A-lines and 512 B-scans.

Hyperreflective foci in neurosensory retina locate between inner nuclear layer (INL) and photoreceptor outer segment (PROS).[24] Thus, segmentation lines were positioned at the superior boundary of INL and inferior boundary of PROS, with RPE layers excluded from segmentation area. Entropy value for each pixel (11.7 μm x 4.4 μm in lateral × axial directions) area was measured by a built-in program (ROCTIA), with moduli values ranging from 0 (fully uniform) to 1 (fully random polarization). The average entropy value for the INL-PROS layer was then calculated and displayed in 2-dimensional *en face* fundus images reconstructed from 3-dimensional data set. To isolate area with high entropy, we established a cutoff based on measurements from healthy controls. The entropy values of these controls were analyzed using the identical method, setting the cutoff at the mean and 2 standard deviations to filter out the background entropy of neurosensory retina. In comparing HRF and hyperentropic foci (HEF), HRF were defined as small, punctiform, and isolated reflective dots blocking signal underneath in OCT images.[12,25] The number and size of HRF and HEF were measured within a 2 mm foveal circle in identical B-scan images acquired by SS-OCT and PS-OCT, with further extraction from *en face* images.[26] To quantify HEF, we calculated "total entropy" as the sum of the entropy values in areas exceeding the cutoff. Additionally, "area aggregate" was determined as the total area surpassing the cutoff.

Statistical Analysis

Data are presented as mean ± SD. Statistical analysis was conducted using JMP Pro software (version 18) (JMP Statistical Discovery LLC). Wilcoxon rank sum test was used to compare 2 groups. In multivariate analysis, generalized linear model was performed. P value < 0.05 was considered as a statistical significance.

Results

Participants

Polarization-sensitive OCT images from 12 eyes of 12 patients with nAMD treated with faricimab were collected. The mean age was 77.4 ± 8.6 years, ranging from 63 to 84 years. Among these patients, 6 were female and 6 were male. Table 1 provides a summary of the epidemiologic and baseline characteristics of the patients.

Table 1. Demographic Baseline Characteristics

Characteristic	Participants (n = 12)
Age, yrs	77.4 ± 8.6
Sex, female/male	6/6
BCVA, logMAR	0.4803 ± 0.3088
CST, μm	355.4 ± 111.2
nAMD type (%)	
MNV1	3 (25%)
MNV2	6 (50%)
MNV3	0 (0%)
PCV	3 (25%)

LogMAR was converted from Snellen visual acuity.
BCVA = best-corrected visual acuity; CST = central subfield thickness; logMAR = logarithm of the minimum angle of resolution; MNV = macular neovascularization; nAMD = neovascular age-related macular degeneration; PCV = polypoidal choroidal neovasculopathy.

Visualization of HRF Using PS-OCT

As the first step, we measured entropy value of INL-PROS in healthy control eyes without any disease conditions to establish a cutoff value for extracting HRF. The measured entropy was 0.0896 ± 0.0270. Based on the mean and 2 standard deviations (0.1436), we defined "0.15" as the cutoff in this study. Next, we have extracted areas exceeding this cutoff from *en face* PS-OCT images. We have defined regions of high entropy value above the cutoff as "hyperentropic foci (HEF)," because there is a possibility that HEF and HRF are not always identical. However, HRF were effectively identified as HEF in the *en face* images of PS-OCT (Fig 2), with 96% of HRF detected as HEF. Interestingly, PS-OCT captured HEF as larger lesions compared with HRF detected by SS-OCT (Fig 3A, B). The significantly larger HEF size was consistently observed both before and after treatment (Fig 3C, D), suggesting that PS-OCT detects not only HRF but also additional surrounding materials.

HEF Quantified by PS-OCT Correlates with Disease Severity

We investigated whether the HEF in neurosensory retina, representing HRF, was associated with baseline characteristics. The sum of the entropy exceeding the cutoff was calculated as the "total HEF," which reflects the total amount of HRF. Multivariate analysis revealed that total HEF, along with sex and nAMD subtype, was significantly associated with baseline BCVA before the treatment (Table 2). The correlation between worse baseline BCVA and higher HEF levels, in contrast to the lack of association with conventional SS-OCT reflectivity.

Faricimab Reduces HEF Lesions in nAMD Eyes

We next examined whether intravitreal administration of faricimab could reduce HEF infiltration in nAMD eyes observed before treatment. Four initial injections of faricimab significantly improved both BCVA and central subfield thickness. Additionally, both the "total HEF" and the "area aggregate" of regions with entropy value above the cutoff were significantly reduced after faricimab treatment (Table 3). Neovascular age-related macular degeneration eyes consistently exhibited lower "total HEF" and "area aggregate" after treatment (Fig 4A), as represented by reconstructed *en face* images in Figures 5 and 6. In contrast, SS-OCT did not show any significant changes in the total reflectivity and the area aggregate of high reflectivity after treatment (Fig 4B). These findings demonstrate that PS-OCT effectively identifies HRF from cross sectional images and that faricimab reduced these lesions alongside exudative changes of nAMD.

Discussion

This study explored the significance of HEF, representing HRF, as an indicator of nAMD severity and the therapeutic efficacy of anti-VEGF treatment. Previous studies have characterized subretinal hyperreflective material and HRF, including the migration of RPE cells, using PS-OCT.[27–29] These reports highlighted the utility of PS-OCT for assessing retinal pathology by identifying high entropy lesions in the macula during nAMD. In the present study, we quantified HRF using entropy values and found that faricimab was sufficient to reduce HRF, supporting its role as a biomarker for both the disease severity and treatment efficacy.

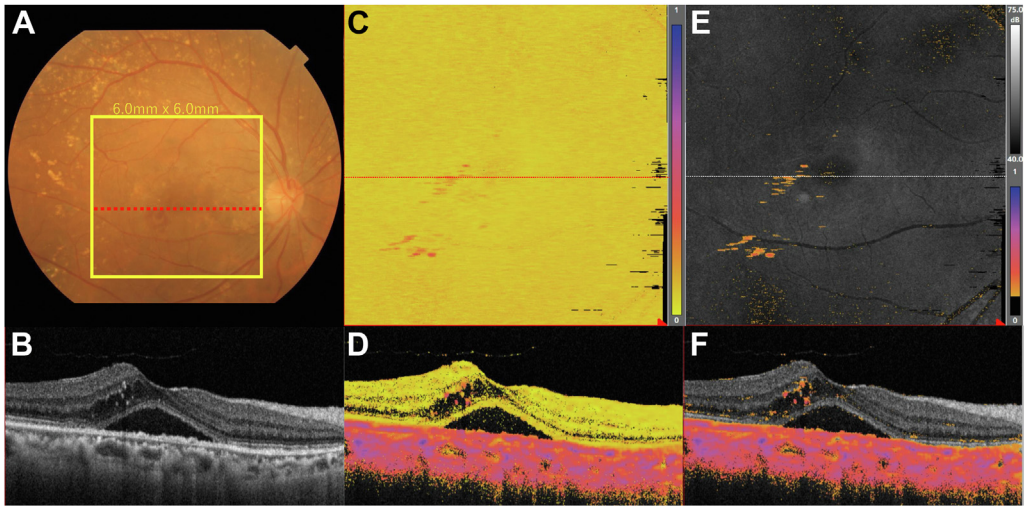


Figure 2. Representative images of neovascular age-related macular degeneration (polypoidal choroidal neovasculopathy [PCV]) eye with hyperreflective foci (HRF) lesions. Fundus photography (**A**) and OCT image (**B**) of PCV presenting with HRF in foveal area. Yellow square indicates the scanned area. The red line in fundus photography and the white line in *en face* polarization-sensitive OCT (PS-OCT) images correspond to B-scan line of OCT. *En face* (**C**) and B-scans (**D**) PS-OCT image of entropy between inner nuclear layer and photoreceptor outer segment were reconstructed from 3-dimensional data set. *En face* (**E**) and B-scan (**F**) PS-OCT image extracting hyperentropic foci using the cutoff entropy value (0.15).

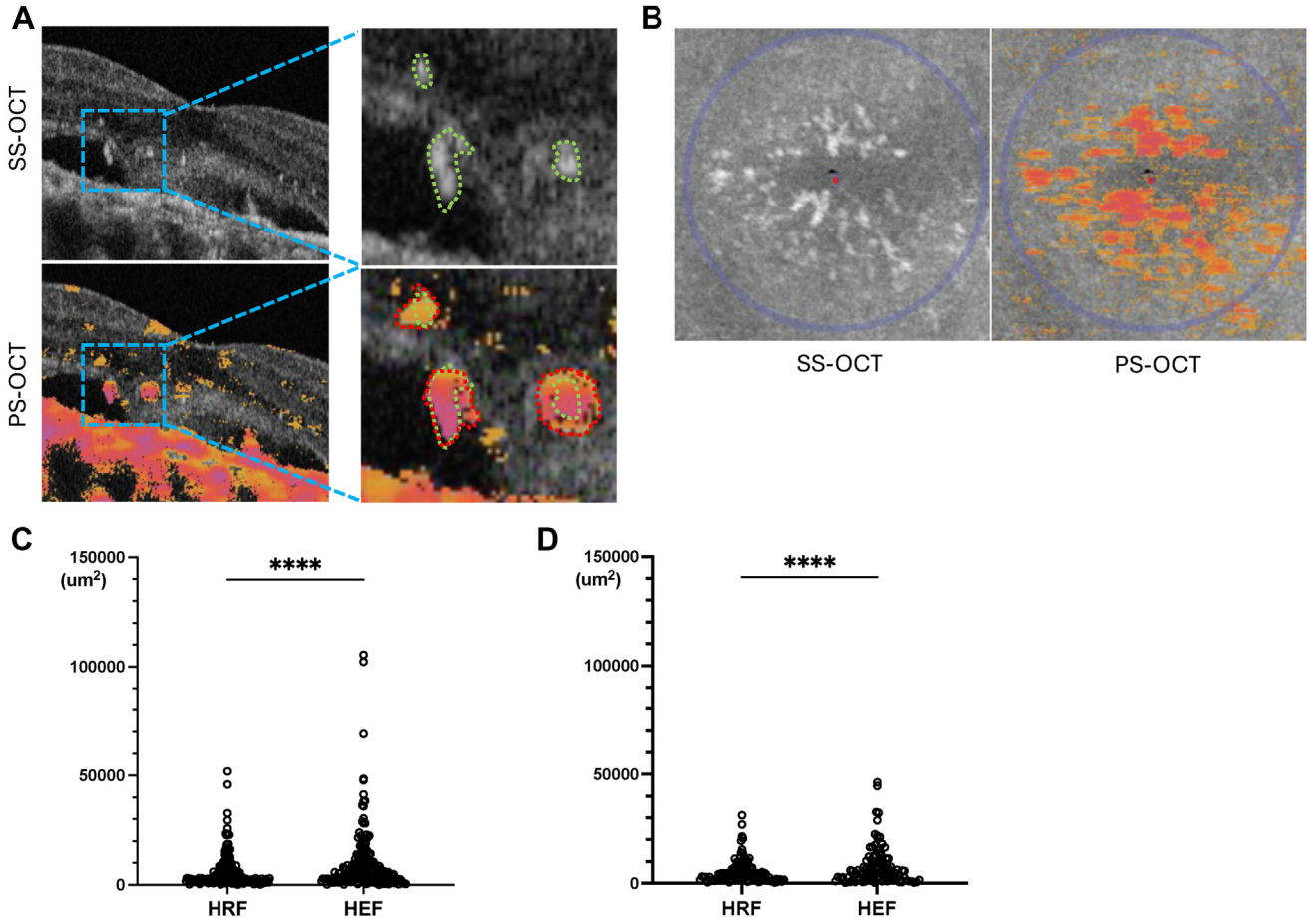


Figure 3. The quantification and comparison of hyperreflective foci (HRF) and hyperentropic foci (HEF). **A,** Swept-source OCT (SS-OCT) and polarization-sensitive OCT (PS-OCT) taken from the identical site. Dash lines indicate the size of HRF an HEF. Note that HEF are consistently larger than HRF. **B,** *En face* images of SS-OCT and PS-OCT in foveal circle of 2 mm diameter. **C,** The size of HRF (n = 208) and HEF (n = 207) before the treatment. **D,** The size of HRF (n = 103) and HEF (n = 100) after the treatment. ****$P < 0.0001$, Wilcoxon rank sum test.

Hyperreflective foci lesions were detected across the neurosensory retina in nAMD, especially in areas with exudative change, as identified by both conventional OCT and PS-OCT. However, conventional OCT struggles to isolate HRF solely based on reflectivity because the signal is comparable to retinal ganglion cell layer, inner plexiform layer, and outer plexiform layer. Moreover, prior studies often relied on manual counting of HRF in conventional OCT images, which introduces potential biases.[12,14] In contrast, PS-OCT measures entropy values, making it a more objective method for HEF quantification, suggesting that PS-OCT is a reliable modality for the quantification of HRF. Our study found no significant changes in reflectivity on conventional OCT before and after anti-VEGF treatment. On the other hand, because HEF is considerably higher compared with ones of any layers of neurosensory retina, cross-sectional PS-OCT easily identified HEF. Moreover, PS-OCT revealed a substantial reduction in HEF by anti-VEGF therapy, confirming its superior ability to isolate and quantify HEF lesions.

Table 2. Multivariate Analysis to Identify the Potential Association among Baseline Visual Acuity and Characteristics

Characteristics	Coefficient	SE	*P* Value
Age, yrs	−0.032069	1.9645323	0.4055
Sex (female)	−1.096512	0.0335439	0.0026
CST	0.0002649	0.3839466	0.8483
nAMD type 1	−3.147091	0.0013841	0.0074
nAMD type 2	2.5087197	0.7709215	0.0004
nAMD PCV	0	—	—
Total HEF	0.0015297	0.0004474	0.0004
OCT reflectivity	1.1666e-7	1.2627e-7	0.3479

CST = central subfield thickness; HEF = hyperentropic foci; nAMD = neovascular age-related macular degeneration; PCV = polypoidal choroidal neovasculopathy; SE = standard error.

Hyperreflective foci is originally defined as discrete, well-circumscribed intraretinal lesions with high reflectivity on conventional OCT, corresponding to hyperpigmentation on fundus photography.[30] They are often found in neurosensory retina with exudative change such as fluid leakage and hemorrhage.[25,31] Hyperreflective foci as well as drusen and subretinal drusenoid deposits predict the

Table 3. Effects of Intravitreal Administration of Faricimab on BCVA, CST, and Entropy Measured by PS-OCT and Reflectivity Measured by SS-OCT

Outcomes	Biomarkers	Before	After	*P* Value
BCVA, logMAR		0.4803 ± 0.3088	0.2648 ± 0.2667	0.0024
CST, μm		355.4 ± 111.2	183.1 ± 44.67	0.0005
PS-OCT	Total HEF (A.U.)	1939 ± 1669	957.2 ± 900.9	0.0049
	Total area (μm^2)	15 063 ± 12 873	7414 ± 6587	0.0068
SS-OCT	Total reflectivity (A.U)	1 8 95 069 ± 2 251 098	1 486 697 ± 2 278 909	0.3394
	Total area (μm^2)	60 066 ± 71 216	47 166 ± 72 161	0.3394

BCVA = best-corrected visual acuity; CST = central subfield thickness; HEF = hyperentropic foci; logMAR = logarithm of the minimum angle of resolution; PS-OCT = polarization-sensitive OCT; SS-OCT = swept-source OCT.

disease progression of late AMD, appearing more frequently at advanced stages of AMD.[12,32] Histologically, HRF are derived from RPE cells, lipid-filled macrophages, microglia, and cellular debris,[7,33,34] which is also detected by PS-OCT as depolarized materials.[18–20] Therefore, intraretinal HEF are considered to be derived from exudates, migrated RPE, and activated immune cells. Smaller HRF (≤30 μm) are considered to represent inflammatory cells, whereas larger lesions correspond to retinal exudates.[25] Immunohistochemically, HRF-related cells express CD38 and CD163 but lack RPE65 and CRALBP, indicating they are transdifferentiated RPE cells or monocytes including macrophages.[33] Hyperreflective foci derived from RPE cells migrates from RPE layer into inner neurosensory retina, whereas microglia infiltrate from inner retinal layers into outer retinal layers.[35] This abnormal cell distribution of RPE cells and inflammatory cells further causes chronic inflammation and angiogenic responses, contributing to disease progression. Additionally, proliferation and migration of HRF were associated with higher incidence of geographic atrophy.[30] In nAMD, HRF correlated with poor initial visual acuity,[36] and some HRF represent exudative changes due to fluid leakage from macular neovascularization. These observations emphasize the

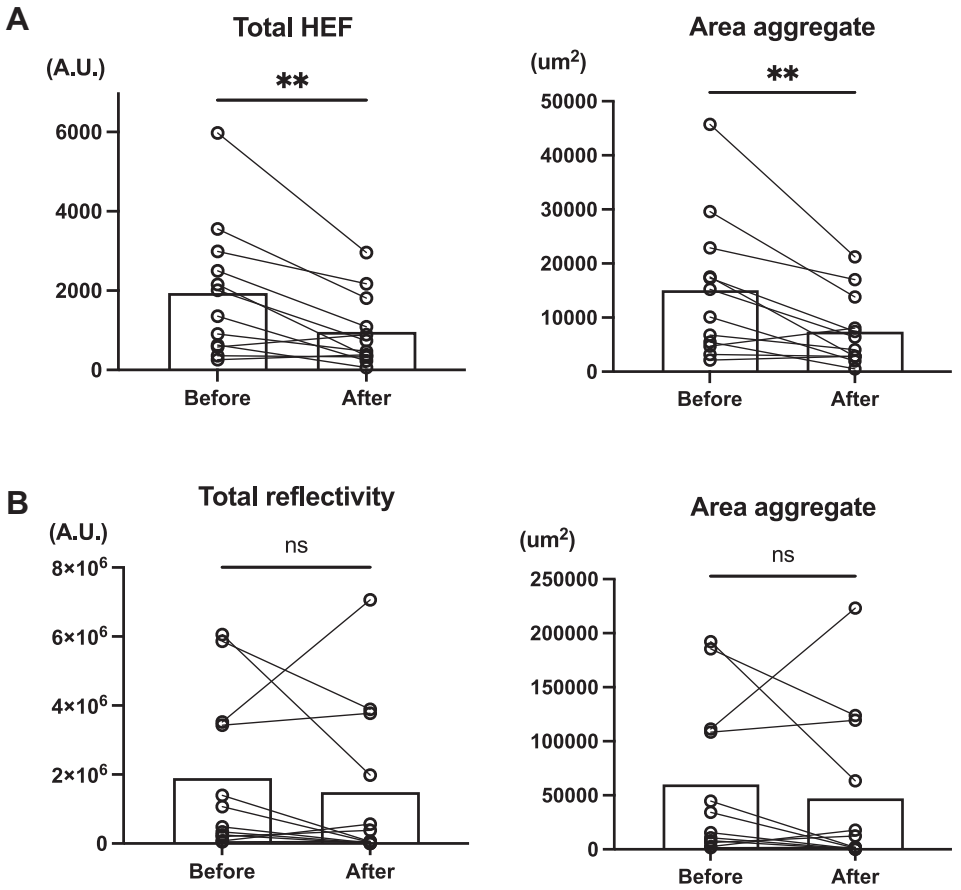


Figure 4. Effects of intravitreal administration of faricimab on entropy measured by polarization-sensitive OCT (**A**) and reflectivity measured by swept-source OCT (**B**). Connecting lines indicate identical eyes before/after treatment. $^{**}P < 0.01$, Wilcoxon rank sum test.

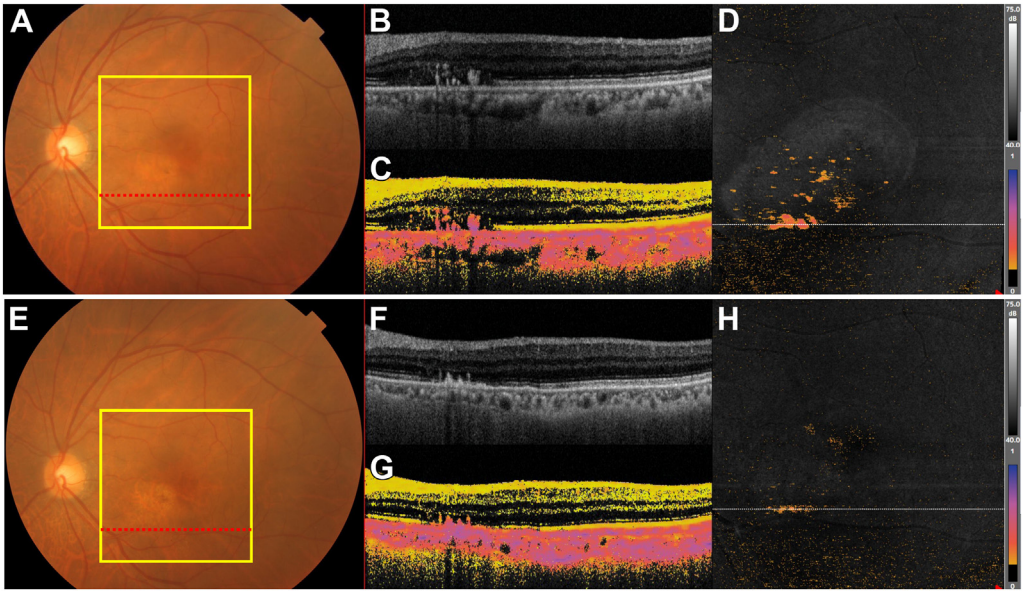


Figure 5. Representative images of macular neovascularization type 1 with hyperreflective foci (HRF) and hyperentropic foci (HEF). Fundus photography (**A, E**), swept-source OCT (**B, F**), polarization-sensitive OCT (PS-OCT) (**C, G**), and *en face* PS-OCT (**D, H**) before (**A–D**) and after (**E–H**) anti-VEGF therapy. Note that HEF corresponding to HRF are reduced by the treatment. Yellow square indicates the scanned area. Red lines in fundus photography and white lines in *en face* PS-OCT images correspond to B-scan line of OCT.

importance of evaluating HRF as a biomarker for disease activity and progression in AMD. Consistent with previous findings, our study found that higher HEF levels were significantly associated with worse baseline BCVA, underscoring HRF quantified by PS-OCT as a reliable biomarker for assessing nAMD disease severity. Notably, 96% of HRF were identified as HEF. Although we could not determine the exact composition of HRF subtypes that was not extracted as HEF, we speculate that certain HRF subtypes may not exhibit high entropy characteristics. Interestingly, HEF were significantly larger than HRF. Although further studies are needed to elucidate the underlying mechanisms, these findings suggest that PS-OCT may detect not only HRF but also additional surrounding materials,

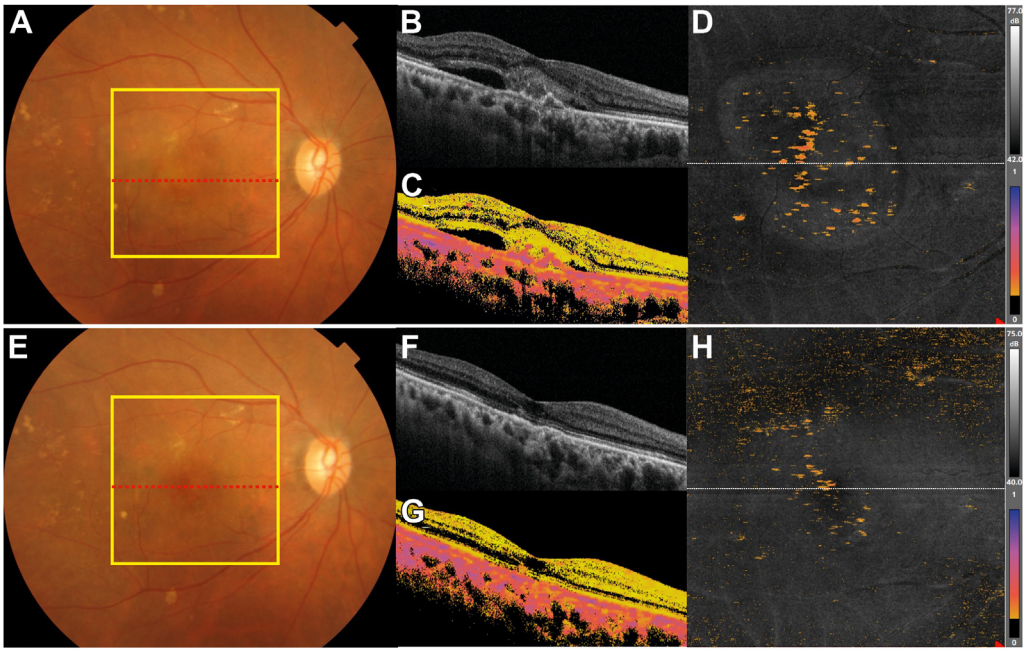


Figure 6. Representative images of macular neovascularization type 2 with hyperreflective foci (HRF) and hyperentropic foci (HEF). Fundus photography (**A, E**), swept-source OCT (**B, F**), polarization-sensitive OCT (PS-OCT) (**C, G**), and *en face* PS-OCT (**D, H**) before (**A–D**) and after (**E–H**) anti-VEGF therapy. Note that HEF regarded are reduced by the treatment. Yellow square indicates the scanned area. Red lines in fundus photography and white lines in *en face* PS-OCT images correspond to B-scan line of OCT.

such as activated immune cells, including microglia, and macrophages.

Our study also revealed that the amount HEF in eyes with nAMD is substantially reduced by anti-VEGF therapy, demonstrating that PS-OCT effectively identifies HRF from cross-sectional images, and that faricimab reduced these lesions alongside exudative changes of nAMD. Anti-VEGF agents reduces intraretinal HRF, which contributes to improved visual outcomes.[36,37] Moreover, residual HRF in the subretinal layer has been associated with worse visual outcomes in nAMD and polypoidal choroidal neovasculopathy.[38] Similar therapeutic effects have been observed in patient with diabetic retinopathy.[39,40] Anti-VEGF therapy improves both visual function and morphological changes, such as reducing serous retinal detachment and intraretinal fluid.[41] Given the availability of multiple treatment options for nAMD, including various anti-VEGF agents and other therapies such as photodynamic therapy, our approach in this study offers a new perspective on evaluating treatment efficacy by focusing on removal of HEF.

This study has several limitations. First, the small sample sizes limit the generalizability of our findings, and future studies with larger cohorts are needed to validate the utility of PS-OCT for HEF quantification. Second, we evaluated only faricimab to minimize potential bias introduced by different anti-VEGF agents. Future research will investigate the efficacy of other anti-VEGF drugs in HEF reduction and compare their effects. Third, although our observations suggest that HRF and HEF are identical, there remains the possibility that HEF identified using PS-OCT does not always correspond to HRF observed in conventional OCT. This discrepancy highlights the need for further investigation into how these imaging modalities correlate precisely. Fourth, this research did not include the longitudinal study, which will enable to further understand the quantitative and spatial changes in HEF after treatment. Finally, we could not determine the specific cellular origin of HEF lesions. As previously noted, HRF can arise from various cell types and pathological processes. Although entropy-based quantification may not allow for precise classification of HEF, PS-OCT could potentially differentiate HEF by retinal layers location, as the cellular composition may vary depending on their position. Future studies using donor eyes and animal models are warranted to better identify the pathological sources of HEF.

In summary, we successfully identified intraretinal HEF using PS-OCT and established a protocol for their quantification. Polarization-sensitive OCT is a valuable imaging modality for evaluating HEF in eyes with nAMD, diabetic retinopathy, and retinal vein occlusion. The quantified HEF detected by PS-OCT serve as a novel biomarker for assessing disease activity, providing insights into the pathogenesis of these retinal diseases.

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Author Contributions:

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Analysis and interpretation: Terao, Aoki, Kitamoto, Totani, Yamanari, Sugiyama, Inoue, Obata, Azuma

Data collection: Aoki, Kitamoto, Totani, Azuma

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Overall responsibility: Terao

Abbreviations and Acronyms:
BCVA = best-corrected visual acuity; **HEF** = hyperentropic foci; **HRF** = hyperreflective foci; **INL** = inner nuclear layer; **nAMD** = neovascular age-related macular degeneration; **PROS** = photoreceptor outer segment; **PS-OCT** = polarization-sensitive OCT; **RPE** = retinal pigment epithelium; **SS-OCT** = swept-source OCT.

Keywords:
Age-related macular degeneration, Polarization-sensitive OCT, Hyperreflective foci, Hyperentropic foci, Anti-VEGF.

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